

SUSTUNO: AI-Driven Predictive Intelligence for Textile Dyeing and Wastewater Treatment

Dr. Sudhakar Karuppusamy

Professor and Head, Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), CMS College of Engineering and Technology, Coimbatore, India

Akshay Sreekumar

UG Scholar, Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), CMS College of Engineering and Technology, Coimbatore, India

Adish S

UG Scholar, Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), CMS College of Engineering and Technology, Coimbatore, India

Laya Chandroth

UG Scholar, Department of Computer Science and Engineering, CMS College of Engineering and Technology, Coimbatore, India

Adwitha K

UG Scholar, Department of Computer Science and Engineering, CMS College of Engineering and Technology, Coimbatore, India

ABSTRACTION:

The textile dyeing industry involves complex combinations of dyes, chemicals, water and process parameters, where variations in production recipes can significantly influence the characteristics of the resulting wastewater. Conventional wastewater treatment systems primarily operate based on the wastewater characteristics observed after generation, creating a gap between upstream production decisions and downstream treatment requirements. This reactive approach limits the ability to anticipate wastewater behaviour and make informed treatment decisions at an early stage. Therefore, an intelligent system capable of connecting textile production decisions with wastewater prediction and treatment intelligence is required.

This paper presents **Sustuno**, an AI and IoT-based predictive intelligence platform designed to establish a continuous connection between textile dyeing operations and effluent treatment processes. The proposed system initially analyses production requirements such as fabric type, colour, batch quantity and process conditions to recommend an optimized dyeing recipe. Based on the selected production parameters, a Machine Learning-based prediction module estimates the expected wastewater characteristics before the wastewater reaches the treatment stage.

The proposed framework consists of multiple functional stages, including production data collection, recipe optimization, wastewater prediction, IoT-based monitoring, expected-versus-actual analysis and ETP decision support. During or after the dyeing process, an IoT sensing system based on ESP32 monitors important wastewater parameters such as pH,

electrical conductivity, turbidity, temperature and flow. The collected real-time measurements are transmitted to the software platform and compared with the predicted wastewater profile. This comparison enables the identification of deviations and abnormal wastewater behaviour.

A significant feature of Sustuno is its approach toward rapid wastewater intelligence. Conventional laboratory analysis of parameters such as **Biochemical Oxygen Demand (BOD)** and **Chemical Oxygen Demand (COD)** can require additional processing and waiting time. Sustuno aims to use real-time sensor parameters and Machine Learning models to provide rapid estimates of such wastewater characteristics, while laboratory measurements can be used for model training, calibration and validation. This approach has the potential to provide earlier information for operational decision-making without replacing established laboratory or regulatory testing procedures.

The platform subsequently converts the predicted and observed wastewater information into **ETP decision support**, enabling operators to understand the changing influent profile and make more informed treatment decisions. Rather than replacing existing ETP infrastructure, Sustuno functions as an intelligence layer that connects production, wastewater monitoring and treatment operations through batch-level traceability.

The effectiveness of the proposed system can be evaluated using prediction error, deviation between predicted and measured parameters, sensor accuracy, response time and consistency of model estimates with laboratory observations. The initial prototype can be further validated through controlled textile dyeing experiments followed by deployment in real textile processing environments. Future development will focus on expanding real-world datasets, improving prediction models, integrating additional wastewater parameters and validating the system across different dyeing processes and textile clusters.

In conclusion, **Sustuno transforms the conventional reactive relationship between textile production and wastewater treatment into a predictive, data-driven decision process.** By linking dyeing recipe optimization, wastewater prediction, real-time IoT monitoring and ETP decision support, the proposed platform provides a foundation for intelligent and sustainable textile processing.